



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

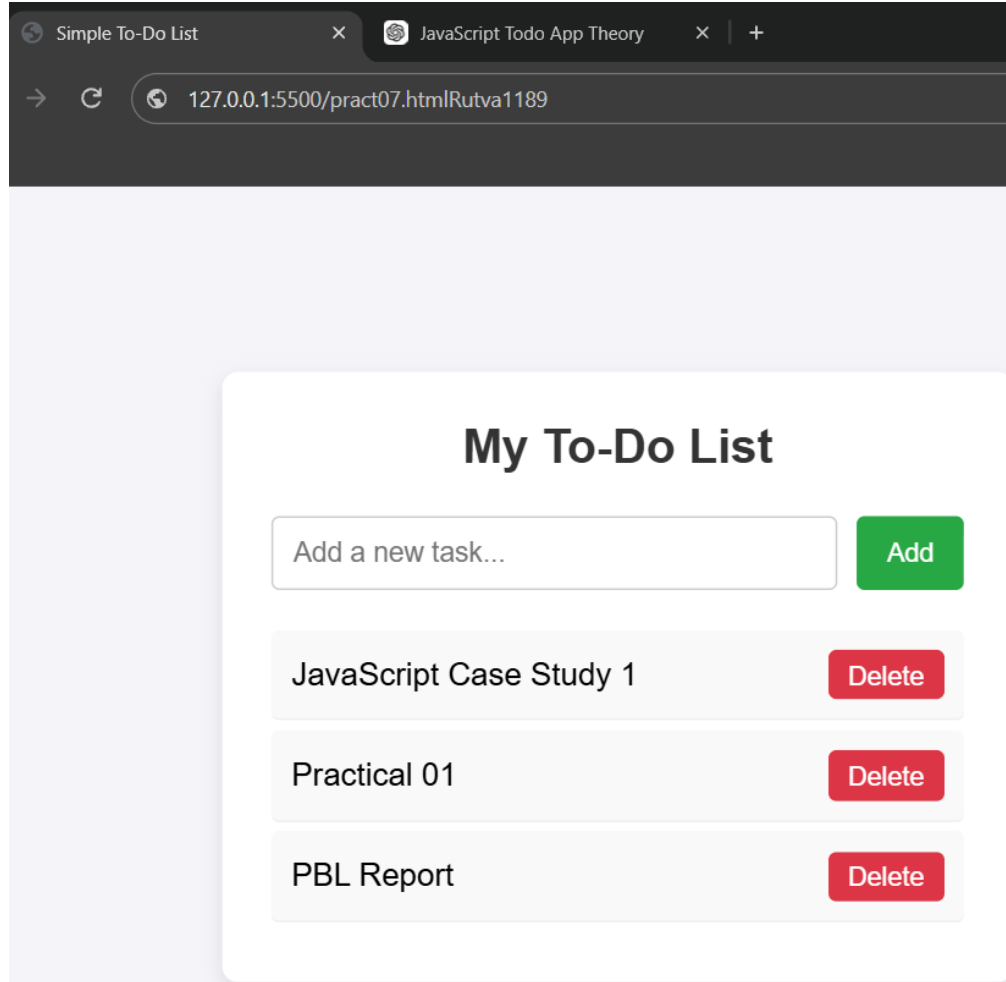
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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment No.: 7

- 1. Title:** Perform DOM traversal and updates; develop a to-do list app with add, edit, and delete features.
- 2. Software/Tools Required:** Visual Studio Code (VS Code), web browser, HTML, JavaScript.
- 3. Theory:** DOM (Document Object Model) traversal is the process of accessing and moving through HTML elements using JavaScript. Methods such as `getElementById()`, `querySelector()`, `parentElement`, `children`, and `nextElementSibling` are used to locate and navigate elements in the DOM. DOM updates allow JavaScript to dynamically change webpage content, attributes, and styles without reloading the page. A **To-Do List application** demonstrates these concepts by allowing users to add new tasks, edit existing tasks, and delete completed or unwanted tasks. JavaScript creates and modifies HTML elements using methods such as `createElement()`, `appendChild()`, `textContent`, and `remove()`. Event listeners such as `click` are used to perform actions when the user interacts with buttons. Thus, DOM traversal and updates make the application interactive and dynamic.



Conclusions: The To-Do List application successfully demonstrates the use of **JavaScript DOM traversal and manipulation**. Tasks can be dynamically added, edited, and deleted based on user actions. This experiment provides a basic understanding of how JavaScript interacts with HTML elements and how event handling can be used to create interactive web applications.



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Case Study No.: 07

1. Case Study Title: To Demonstrate the Events Handling in JavaScript

- Access and manipulate form elements using JavaScript.
- Validate user input (e.g., email, empty fields).
- Use various form events (submit, change, focus). Create the following HTML form and use required JavaScript and events in a given form.

2. Case Study Program Code:

```
cs07.html > html
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Event Handling Form</title>
5
6      <style>
7          body {
8              font-family: Arial, sans-serif;
9              background-color: #f2f2f2;
10             padding: 30px;
11         }
12
13         .container {
14             width: 450px;
15             margin: auto;
16             background-color: white;
17             padding: 25px;
18             border-radius: 10px;
19         }
20
21         h1 {
22             text-align: center;
23         }
24
25         label {
26             display: block;
27             margin-top: 12px;
28             font-weight: bold;
29         }
30
31         input, select {
32             width: 95%;
33             padding: 9px;
34             margin-top: 5px;
35         }
36
37         button {
38             margin-top: 20px;
39             padding: 10px 25px;
40             background-color: black;
41             color: white;
42             border: none;
43             cursor: pointer;
44         }
45     </style>
46 </head>
47 <body>
48     <div class="container">
49         <h1>Student Registration Form</h1>
50
51         <form id="myForm">
52             <label>Full Name</label>
53             <input type="text" id="name" placeholder="Enter your name">
54             <p id="nameError" class="error"></p>
55
56             <label>Email</label>
57             <input type="text" id="email" placeholder="Enter your email">
58             <p id="emailError" class="error"></p>
59
60             <label>Course</label>
61             <select id="course">
62                 <option value=""-- Select Course --</option>
63                 <option value="CSE">Computer Science</option>
64                 <option value="DS">Data Science</option>
65                 <option value="IT">Information Technology</option>
66             </select>
67
68             <label>Phone Number</label>
69             <input type="text" id="phone" placeholder="Enter 10 digit number">
70             <p id="phoneError" class="error"></p>
71         </form>
72     </div>
73 </body>
74 </html>
```

```
46     button:hover {
47         background-color: #555;
48     }
49
50     .error {
51         color: red;
52         font-size: 13px;
53     }
54
55     #message {
56         font-weight: bold;
57         margin-top: 15px;
58     }
59 </style>
60 </head>
61 <body>
62     <div class="container">
63         <h1>Student Registration Form</h1>
64
65         <form id="myForm">
66             <label>Full Name</label>
67             <input type="text" id="name" placeholder="Enter your name">
68             <p id="nameError" class="error"></p>
69
70             <label>Email</label>
71             <input type="text" id="email" placeholder="Enter your email">
72             <p id="emailError" class="error"></p>
73
74             <label>Course</label>
75             <select id="course">
76                 <option value=""-- Select Course --</option>
77                 <option value="CSE">Computer Science</option>
78                 <option value="DS">Data Science</option>
79                 <option value="IT">Information Technology</option>
80             </select>
81
82             <label>Phone Number</label>
83             <input type="text" id="phone" placeholder="Enter 10 digit number">
84             <p id="phoneError" class="error"></p>
85         </form>
86     </div>
87 </body>
88 </html>
```

```

88     <p id="phoneError" class="error"></p>
89
90     <button type="submit">Submit</button>
91
92     <p id="message"></p>
93
94 </form>
95
96 </div>
97
98
99 <script>
100     // Access form elements
101     let form = document.getElementById("myForm");
102     let name = document.getElementById("name");
103     let email = document.getElementById("email");
104     let course = document.getElementById("course");
105     let phone = document.getElementById("phone");
106
107
108     // Focus event
109     name.addEventListener("focus", function() {
110         name.style.backgroundColor = "#eeeeee";
111     });
112
113
114     // Change event
115     course.addEventListener("change", function() {
116         document.getElementById("message").innerHTML =
117             "Selected Course: " + course.value;
118     });
119
120
121     // Email validation using input event
122     email.addEventListener("input", function() {
123         let pattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
124
125         if (!pattern.test(email.value)) {
126

```

```

129             document.getElementById("emailError").innerHTML =
130                 "Enter a valid email.";
131         }
132     } else {
133         document.getElementById("emailError").innerHTML = "";
134     }
135 });
136
137
138 // Phone validation using input event
139 phone.addEventListener("input", function() {
140
141     let pattern = /^[0-9]{10}$/;
142
143     if (!pattern.test(phone.value)) {
144         document.getElementById("phoneError").innerHTML =
145             "Phone number must contain 10 digits.";
146     }
147     else {
148         document.getElementById("phoneError").innerHTML = "";
149     }
150 });
151
152
153 // Submit event
154 form.addEventListener("submit", function(event) {
155
156     // Prevent actual form submission
157     event.preventDefault();
158
159     let valid = true;
160
161     // Name validation
162     if (name.value.trim() === "") {
163
164         document.getElementById("nameError").innerHTML =
165             "Name cannot be empty.";
166         valid = false;
167     }
168

```

```

182         valid = false;
183     }
184
185
186     // Course validation
187     if (course.value === "") {
188
189         document.getElementById("message").innerHTML =
190             "Please select a course.";
191         valid = false;
192     }
193
194
195     // Phone validation
196     let phonePattern = /^[0-9]{10}$/;
197
198     if (!phonePattern.test(phone.value)) {
199
200         document.getElementById("phoneError").innerHTML =
201             "Enter a valid 10-digit phone number.";
202         valid = false;
203     }
204
205
206     // Final result
207     if (valid) {
208
209         document.getElementById("message").innerHTML =
210             "Form submitted successfully!";
211     }
212
213
214 });
215
216
217 </script>
218
219 </body>
220 </html>

```

4. Output:

Student Registration Form

Full Name
Rutva Sawarkar

Email
rutva@gmail.com

Course
Computer Science

Phone Number
9307367867

Submit

Form submitted successfully!

Student Registration Form

Full Name
Rutva Sawarkar

Email
rutva gmail.com
Enter a valid email.

Course
-- Select Course --

Phone Number
11
Enter a valid 10-digit phone number.

Submit

Please select a course.

4. Result / Conclusion: The HTML form was successfully created using JavaScript event handling. The form elements were accessed and validated using JavaScript, and focus, change, and submit events were successfully implemented.